Smart India Hackathon 2023

PROBLEM STATEMENT

Ministry/Organisation : Ministry of Jal Shakti
PS Code : SIH1289
Problem Statement Title : Projection of the extent of inundation corresponding to the forecasts of flood levels in a river.

Team Name : Aquarius
Team Leader Name : Agneesh Dasgupta
Institute Code (AISHE) : U-0055
Institute Name : National Institute of Technology, Silchar
Theme Name : Disaster Management

Our Solution

Solution Description :

- We proposed a development of **AI** based **GIS system** that leverage the past historical imagery (Sources: Sentinel-1 SAR datasets) to generate new inundation map. This is done in three stages:

- **Stage 1:** In the first stage, flood inundation extent was estimated, this is done with the help of two models:

  An **U-NET** based image segmentation model to classify **dry terrain**, **permanent water bodies** and flood **inundated areas** in historical imagery data. This model is used as a data preparation strategy for the second model.

  A **Encoder-Decoder** architecture based model to generate imagery of projection of flood inundations considering **segmented imagery** maps (output of UNET model) and **water level predictions** (Source: CWC website).

- **Stage 2:** In the second stage, the generated flood inundation prediction imagery is used along with **Digital Elevation Model (DEM)** to estimate flood water depth map.
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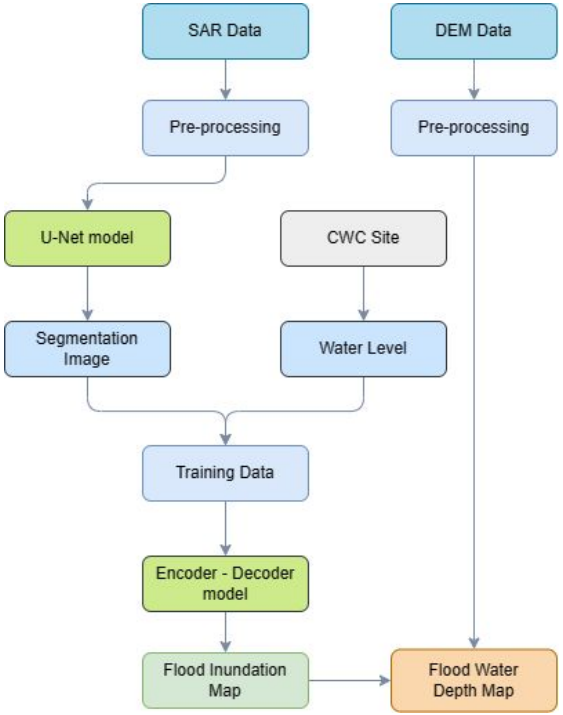
- **Stage 3:** The generated flood water depth map is then plotted into an GIS application using **ArcGIS**.



Technology Stack:

- **Web Interface :** HTML, CSS, JavaScript, React/Next, Django
- **Storage :** PostgreSql
- **GIS Software :** ArcGIS JS SDK
- **Machine learning:** Tensorflow, keras, OpenCV

Architecture & Use Cases

Full Documentation: https://bit.ly/FloodSafeGIS-Documentation

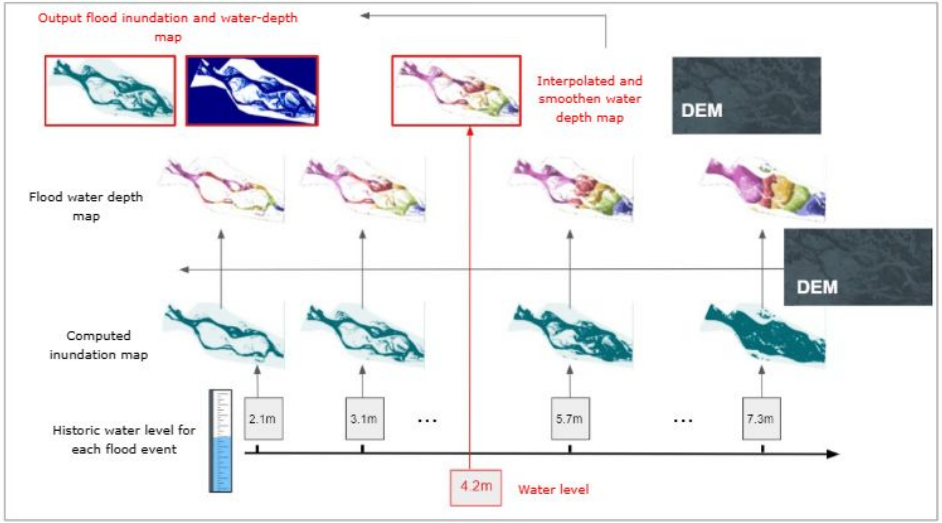
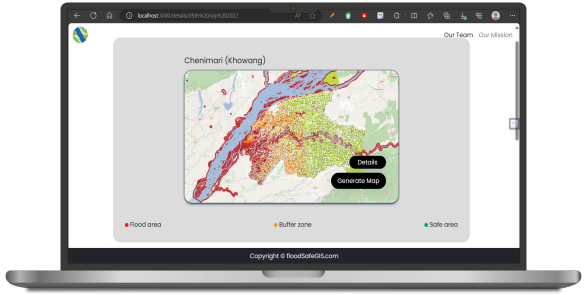


Fig - Solution model

- In the event of a flood, it can provide accurate **flood extent maps**, allowing authorities to identify vulnerable regions and plan evacuation and relief efforts strategically.
- By using the **CWC flood forecast**, it can predict which areas will be affected and estimate the **water depth**.
- It can identify **infrastructures**, **populations affected** and **high-risk areas** which can be used by authorities to prioritize mitigation efforts.
- An **alert system** through **Cell Broadcasting System** to notify residents, emergency services, and government agencies about potential flooding risks.



Prototype: https://floodsafegis.onrender.com

Team Member Details

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Team Mentor 1 Name: Dr. Partha Pakray

Academic AI/ML 9+ Years

Team Mentor 2 Name: Dr. Wasim Arif

Academic Signal Processing/AI/ML 9+ Years